

CS 142: Program design & methodology

Spring 2025

Professor: [David Bunde](#), SMC E-203, 341-7479, dbunde@knox.edu

Office hours: Whenever my office door is open; email for an appointment.

Website: <http://courses.knox.edu/cs142>

Generally, the best way to reach me is via email.

TAs:

Chuyen Nguyen, chtnghuyen@knox.edu, Office hours: 3-5pm Tues under the whale

Saharsh Shrestha, sbshrestha@knox.edu, Office hours: 4-5pm Mon under the whale

Course overview

In this class, you will continue to develop your programming skills and begin to consider program design and efficiency. Central to these concerns is the choice of data structures, the ways your program organizes information. We will see that choosing the right data structures can greatly improve the speed of your program. We will study a variety of data structures that support different operations and also different ways in which they can be implemented to achieve important performance goals. Our focus will be to understand how to select the appropriate data structure type and implementation for each specific application.

Meetings

Our scheduled lecture times are Monday, Wednesday, and Friday during 3rd period (10:40-11:50am) in SMC A-201. We also have a lab scheduled during 3rd period on Thursday in the Stellyes Lab (SMC E-016).

Our midterm exam will be held on Monday 4/21.

Resources

I will use [Google Classroom](#) to post copies of the slides, assignments, and announcements there. I'll invite everyone to our class; if I somehow miss doing this, please send me an email.

This class has a optional textbook, "Data structures and abstractions with Java" (5th ed) by Carrano and Henry. You are not required to get a copy of this book, but it is the book from which I designed most of the lectures. Thus, it may be useful if you want a written source of information on the course material.

You will need to get a Clicker (i>clicker+; available at the bookstore) for this course. This is a radio device that lets you "vote" a multiple-choice answer (A-E). (You can also use the app version, but students have had some trouble with that version in the past.) We'll use it as part of the Peer Instruction pedagogy.

Peer Instruction

This class will be taught using Peer Instruction, in which the lecture is periodically interrupted with multiple choice questions that you answer and then discuss with your peers. (These questions are graded for participation so don't worry if you can't answer them all.) Peer Instruction has been shown to improve your learning; you learn more by explaining your ideas to others and hearing their explanations than by just listening to me.

To get the benefits of Peer Instruction, you need to actively participate. Attempt to correctly answer the questions. Be sure to explain your reasoning and listen to the reasoning of other students when you're in your group. This can be uncomfortable since you'll have to admit when you don't understand something, but this is an opportunity to get clarifications from your peers.

Succeeding in this course

Expect to work for this class. This is one of three classes that make up a full-time schedule so you should spend ~13 1/3 hours/week on it.

The best way to spend your time on the course is to practice. See if you can implement methods for the data structures we talk about in class. Start without looking at other resources, referring to your notes or online resources only if you get stuck; you don't truly understand what we're doing in class unless you can apply it yourself. I also highly recommend getting programming practice. You can expand the programs we do for class or write your own. The key thing is that you spend time thinking about and writing code. (Remember how long it took you to become a proficient writer. Learning to program is every bit as much a process and requires every bit as much practice.)

If you find yourself struggling in this course, get help right away. The term will move quickly and concepts in this course build on one another so you can't afford to get too far behind. Please reach out if you have a question - I'm happy to talk with you if you come by my office and am generally responsive to questions sent by email. You can also get help from the [Red Room SMC tutors](#).

Tentative schedule

The following is a tentative schedule of what we'll cover each week. This instance of the class is being significantly redesigned so the schedule is more tentative than normal. For what we've actually done, see the [Lectures](#) page and Google Classroom.

Week 1: Review of Java. Abstract Data Types (ADTs). List ADT. Generics.

Week 2: Testing. JUnit. Exceptions.

Week 3: Linked memory. Iterators. Running time. Big-O notation.

Week 4: StringBuilder. Stack and Queue ADTs. Design problems.

Week 5: Comparators. Simple sorting algorithms. Recursion. Recursive sorting. **Midterm Exam.**

Week 6: Multi-threaded computing. Fork-join algorithms. Map and Set ADTs. Binary search.

Week 7: Trees and Binary Search Trees (BSTs). Tree traversals. Priority Queue ADT.

Week 8: Bucket/radix sort. Hashing.

Week 9: Graphs. Graph algorithms.

Week 10: Review.

Assignments

This class will have the following types of assignments:

- **Clicker questions.** The in-class questions will be graded for participation only. For full credit, you need to participate on at least 85% of the days with clicker questions. Credit will be reduced proportionally if you miss more than this.
- **Lab assignments.** During lab periods, you will have an assignment sheet to work on while a TA and I circulate to answer questions. The lab period is time for you to practice new material with assistance when needed. Because of this, you will be graded for attending and actively participating in lab, but not for completion of the assignments. Lab assignments are also important because homework assignments are often based on them.
- **Homework.** Assignments for this class will feature a mixture of programming and writing. Most assignments will be fairly short, but some longer assignments may also be given.
- **Exams.** This course will have one midterm exam during the term (on 4/21) and a final exam during the exam period. Both will be open-note, in-class exams with problems similar to the homeworks. The final will be cumulative, but emphasize material since the midterm.

Be aware that I draw my examples for lecture from student assignments. Don't be surprised if something you submit ends up projected on the screen during class. The purpose of this is to demonstrate important concepts and to provide material for discussion, not to embarrass you. I will endeavor to use these examples anonymously by removing names and any other identifying features.

Late work

Exams: The exams will be due at the end of the exam period except as scheduled to meet academic accommodations. A makeup exam will be available if necessary for students with conflicts; these must be declared to me as soon as possible and definitely before the exam period.

HW: Homeworks will be due at midnight. Everyone has an automatic extension for 8 hours (i.e. until 8am the next morning). I may release homework solutions or discuss them in class at any time after the automatic extension, after which those problems cannot be submitted. Otherwise, homework submitted up to 24 hours after the automatic extension incurs a 10% penalty and work submitted between 24 hours after the automatic extension and 2 weeks after the extended deadline incurs a 20% penalty. No submissions will be accepted more than two weeks late without exceptional circumstances. In addition, no homework will be accepted after the end of the first reading day (5/30).

In recognition that sometimes "life happens" and gets in the way of the timely completion of work, you get two "late days" that may be used on homework assignments during the term. Each allows you to reduce the late penalty on an assignment by 10%. They can be used for the same assignment (completely removing the penalty on that assignment) or different assignments (reducing the penalty by 10% on each). Note that the late days do not allow you to get points on assignments completed after two weeks from the extended deadline.

Individual extensions beyond the automatic extensions and these two late days may be granted, but require extraordinary circumstances. "Extraordinary circumstances" are events more serious than things that happen (you have other classes, you get sick, you have a fieldtrip for another class, etc). I have in mind serious injury or illness (you go to the hospital or are incapacitated for an extended time), someone's death, etc.

Labs: If you cannot attend a lab period, let me know ahead of time. If it's an excused absense (sickness, athletic competition, etc), then you'll be allowed to make up the attendance credit by working on the lab for 70 minutes and emailing me your code at that time. This must normally be completed within a week of the lab (i.e. before the next lab) and no later than the end of the first reading day, though extensions may be given for extraordinary circumstances.

Flunk day

If flunk day falls on a day an assignment is due or the day before, that assignment is extended by one day. Flunk day also acts as a free "late day" and does not count toward the 2 weeks during which assignments must be completed.

Extra credit

I assign extra credit problems infrequently, but I will give extra credit for attending colloquium talks sponsored by the CS department (possibly other departments, depending on the talk) and submitting a short writeup about the talk. This offer applies to any qualifying talk for which you are not otherwise getting credit; you should not submit a writeup to me and also for another class.

In order to get credit for a talk, submit a quick writeup (~1 page) to me within two weeks of the talk (you need to remember the talk...). DO NOT just summarize the talk; I want to hear what you thought about it or how it relates to other things you've seen/heard. While your writeup doesn't have to be brilliant prose, I do expect you to write thoughtfully and correctly and I may return your writeup for revisions before giving you credit.

Policy on Collaboration

Exams in this class must be completed individually. You are allowed to talk with others about the homework and are encouraged to work together in small groups (2-3 students). **Even when working together you must write your submission yourself and understand it completely.** In addition, you must do two things:

1. First of all, you must acknowledge all collaborators by adding a note to your solution describing who helped you and how. Get in the habit of doing this as soon as you talk.
2. Secondly, you must avoid "collaborations" where one person gives the solution to another. Never share your code with someone who hasn't already finished the assignment. When working together, each person should contribute ideas. If one person is helping another, the helper can explain concepts and guide the "helpee" toward a solution, but cannot tell them how to solve the problem. The helper can help debug provided they have already finished the assignment and help the helpee identify the bug rather than fixing it for them. In all collaborations, everyone is expected to understand the entire solution and each individual must write up their solution separately.

The same policies apply to getting help from inanimate sources, including AI systems. If you use a source other than our textbook or other course materials, you must cite this source. As in collaborations with another person, you are expected to write your own solution; do not copy text or code from any source. You may look at other sources for examples or conceptual help, but not actual solutions. Some problems assigned for this class may have been used previously, but you are expected not to deliberately seek solutions. (If you come across one accidentally, let me know and I will give you a modified assignment.)

Please feel free to contact me if you have any questions about this policy.

Accommodations

Students who qualify for accommodations in this class are encouraged to contact Stephanie Grimes (sgrimes@knox.edu) in the Office of Disability Support Services as soon as possible to ensure that such accommodations are implemented in a timely fashion. This is particularly important for the midterm; if you need extra time or other accommodations, make sure that this is documented and shared with me before the exam.

Grading

At the end of the term, all of your work will combined into an overall course percentage based on the following weights:

Clicker questions	10%
Lab participation	5%
Homework	40%
Midterm exam	20%
Final	25%

The range of scores mapping to each grade will be determined at the end of the term. It will not be stricter than the "standard" 90/80/70/60 and may be more generous. In general, spend time learning the material rather than worrying about your grade, but feel free to talk with me if you are concerned about it.